

JEE-Main-07-04-2025 (Memory Based)
[MORNING SHIFT]
Physics

Question: The dimensions of a physical quantity $\epsilon_0 \frac{d\phi_E}{dt}$ are similar to [Symbols have their usual meanings]

Options:

- (a) Electric current
- (b) Electric field
- (c) Electric flux
- (d) Electric charge

Answer: (a)

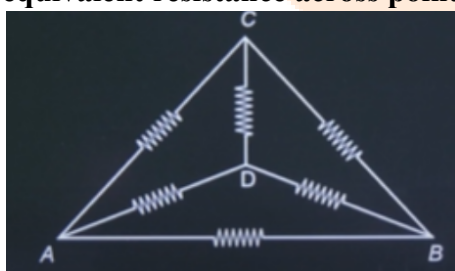
Question: In a resonance tube closed at one end. Resonance is obtained at length $l_1 = 120$ cm and $l_2 = 200$ cm. If $V_s = 340$ m/s. Find frequency of sound is $\frac{100\alpha}{8}$ Hz then $\alpha =$

Options:

- (a) 12
- (b) 15
- (c) 25
- (d) 17

Answer: (d)

Question: Six resistors of resistance R each, are connected as shown in figure. Find equivalent resistance across points A and B.



Options:

- (a) $R/2$
- (b) $R/3$
- (c) $2R/3$
- (d) $3R/2$

Answer (a)

Question: 2 plane polarized light waves combine at certain point whose “E” components are

$$E_1 = E_0 \sin \omega t, E_2 = E_0 \sin(\omega t + \pi/3)$$

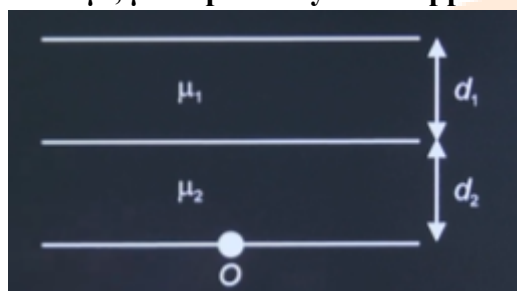
Find the Amplitude of Resultant wave.

Options:

- (a) E_0
- (b) $0.9 E_0$
- (c) $1.7 E_0$
- (d) $3.4 E_0$

Answer: (c)

Question: An object is placed below two parallel layers of thickness d_1 , d_2 are refractive index μ_1 , μ_2 respectively. Find apparent depth of the object.



Options:

- (a) $\frac{d_1 \mu_2 - d_2 \mu_1}{\mu_1 \mu_2}$
- (b) $\frac{d_1 \mu_2 + d_2 \mu_1}{\mu_1 \mu_2}$
- (c) $\frac{d_1 \mu_1 + d_2 \mu_2}{\mu_1 \mu_2}$
- (d) $\frac{d_1 \mu_1 - d_2 \mu_2}{\mu_1 \mu_2}$

Answer: (b)

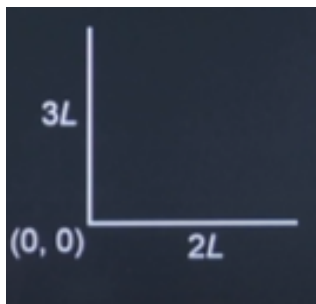
Question: A lens of focal length 20 cm in air is made of glass with refractive index 1.6. What is its focal length when it is immersed in a liquid of refractive index 1.8?

Options:

- (a) -36 cm
- (b) -72 cm
- (c) -60 cm
- (d) -108 cm

Answer: (d)

Question: A rod of length $5L$ is converted in L-shape as shown. Find the position of its center of mass w.r.t origin.



Options:

- (a) $\left(L, \frac{3L}{2}\right)$
- (b) $\left(\frac{9L}{12}, \frac{5L}{12}\right)$
- (c) $\left(\frac{2L}{5}, \frac{9L}{10}\right)$
- (d) $\left(\frac{9L}{10}, \frac{4L}{5}\right)$

Answer: (c)

Question: A composite sound wave is represented by $y = A \cos \omega t \cdot \cos \omega' t$. The observed beat frequency is

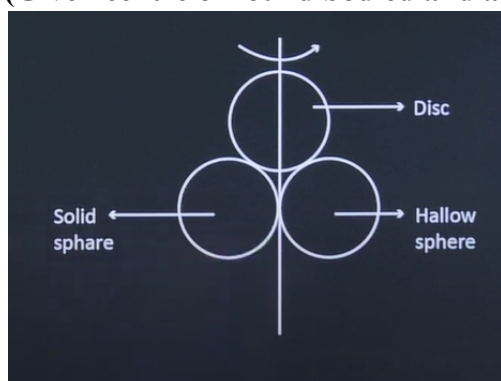
Options:

- (a) $\frac{\omega - \omega'}{2\pi}$
- (b) $\frac{\omega - \omega'}{\pi}$
- (c) $\frac{\omega'}{2\pi}$
- (d) $\frac{\omega'}{\pi}$

Answer: (d)

Question: MOI of disc about central axis perpendicular to surface is I then moment of inertia of given assembly is, where each round object is of same mass and same radius.

(Given centre of round bodied and axis are planar).



Options:

- (a) $\frac{89}{15}I$
- (b) $\frac{79}{17}I$
- (c) $\frac{199}{30}I$
- (d) $\frac{209}{32}I$

Answer: (c)

Question: Two convex lenses of focal length 30 cm and 10 cm are kept 10 cm apart. Principal axis of the lenses is common. Find equivalent power of the lens system.

Options:

- (a) 5 D
- (b) 10 D
- (c) 6 D
- (d) 13.33 D

Answer: (b)

Question: Two rods whose lengths are in ratio of 1 : 3 and of diameter are in ratio of 2 : 1, then ratio of elongations of rod if force applied and material of rods are same

Options:

- (a) 1 : 12
- (b) 1 : 3
- (c) 4 : 1
- (d) 1 : 1

Answer: (a)

$$i = 5\sqrt{2} + 10 \cos\left(650\pi t + \frac{\pi}{6}\right) \text{ Amp.}$$

Question: AC current is represented by
of the current is

The rms value

Options:

- (a) $5\sqrt{2}$ Amp
- (b) 50 Amp

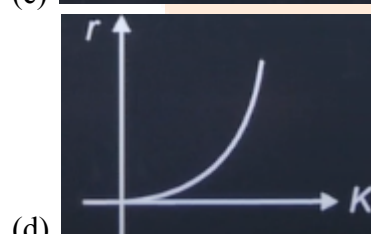
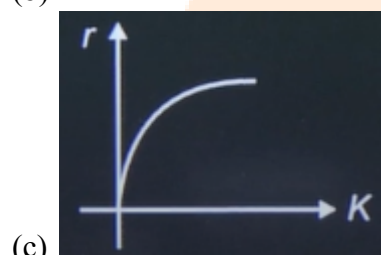
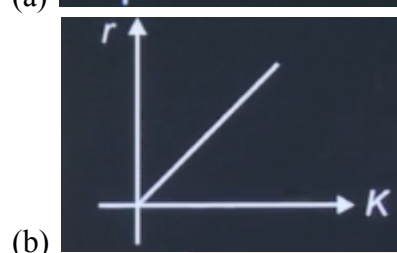
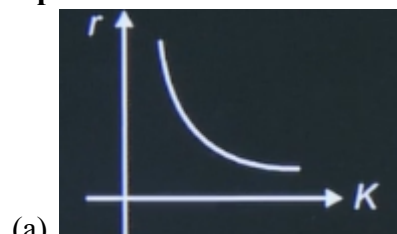
(c) 100 Amp

(d) 10 Amp

Answer: (d)

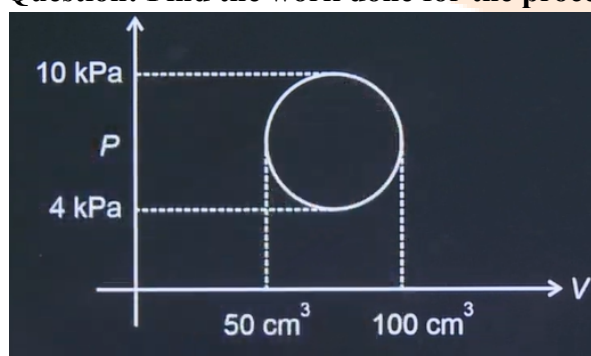
Question: A charge particle moves in circular path in uniform magnetic field. The graph radius of circular path versus its kinetic energy is best and represented by

Options:



Answer: (c)

Question: Find the work done for the process shown in figure.



Options:

(a) $\frac{3\pi}{40}$ J

(b) $\frac{\pi}{20} \text{ J}$

(c) $\frac{\pi}{10} \text{ J}$

(d) $\frac{3\pi}{20} \text{ J}$

Answer: (a)

Question: Match the two columns.

	Column-I		List-II
a.	Monoatomic gas	(i)	$\gamma = 7/5$
b.	Diatomic rigid gas	(ii)	$\gamma = 4/3$
c.	Diatomic non-rigid	(iii)	$\gamma = 5/3$
d.	Triatomic non-linear rigid	(iv)	$\gamma = 9/7$

Options:

(a) a(i), b(iii), c(ii), d(iv)

(b) a(iii), b(i), c(iv), d(ii)

(c) a(ii), b(iv), c(i), d(ii)

(d) a(iii), b(iv), c(i), d(ii)

Answer: (b)

Question: Let λ_1 be largest wavelength of Lyman series for hydrogen atom and λ_2 be

largest wavelength of Balmer series then $\frac{\lambda_1}{\lambda_2}$ is

Options:

(a) $5/27$

(b) $1/4$

(c) $3/4$

(d) $4/9$

Answer: (a)

Question: Two bodies projected with same initial velocities from same point with angle $45 + \alpha$ and $45 - \alpha$. The ratio of their Time of Flights is

Options:

(a) 1

(b) $\frac{1 + \tan \alpha}{1 - \tan \alpha}$

(c) $\frac{1 + \sin 2\alpha}{1 - \sin 2\alpha}$

$$(d) \frac{1 - \tan \alpha}{1 + \tan \alpha}$$

Answer: (b)

Question: Two particles A and B are projected from a point on ground with same speed at angles $45^\circ + \theta$ and $45^\circ - \theta$ with horizontal. If the times of flight are T_A and T_B

respectively. Then $\frac{T_A}{T_B}$ is equal to

Options:

- (a) $\frac{1 + \tan \theta}{1 - \tan \theta}$
 (b) $2 \tan \theta$
 (c) $\tan 2\theta$
 (d) $\frac{1 - \tan \theta}{1 + \tan \theta}$

Answer: (a)

Question: Two plane polarised light combine at certain point whose electric field components are

$$E_1 = E_0 \sin \omega t \hat{i}$$

$$E_2 = E_0 \sin \left(\omega t + \frac{\pi}{3} \right) \hat{i}$$

Find the amplitude of electric field of resultant wave.

Options:

- (a) $0.9 E_0$
 (b) E_0
 (c) $0.7 E_0$
 (d) $3.7 E_0$

Answer: (c)

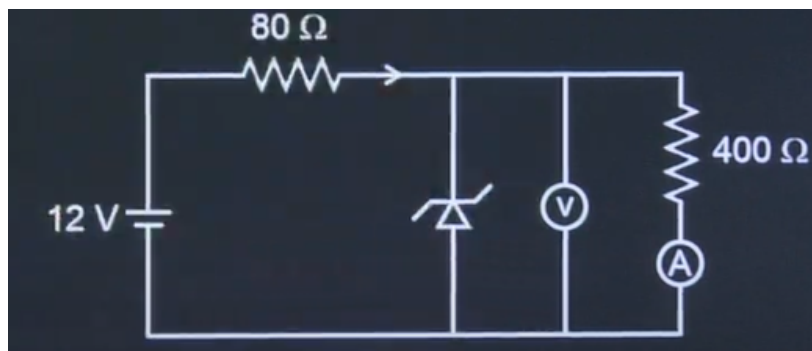
Question: A variable force $\vec{F} = 2t\hat{i} + 3t^2\hat{j}$ acts on a particles of mass 1 kg, which is at rest at $t = 0$. Find the power supplied as a function of time.

Options:

- (a) $2t^3 + 3t^5$
 (b) $t^3 + 4t^5$
 (c) $t^3 + 4t^3$
 (d) $t^3 + 5t^4$

Answer: (a)

Question: Find the current through the ammeter for the circuit shown in figure.
 (Reading voltmeter = 4V)



Options:

- (a) 1 mA
- (b) 7 mA
- (c) 10 mA
- (d) 3 mA

Answer: (c)

Question: Find the dimensions of the physical quantity $\frac{2B^2\mu_0}{\epsilon_0}$. (Symbols have their usual meanings)

Options:

- (a) $M^4L^{-2}T^{-4}A^{-1}$
- (b) $M^4L^4T^{-10}A^{-6}$
- (c) $M^{-1}L^{-2}T^{-3}A^{-1}$
- (d) $M^{-1}L^{-4}A^{-1}$

Answer: (b)

JEE-Main-07-04-2025 (Memory Based)
[MORNING SHIFT]
Chemistry

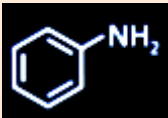
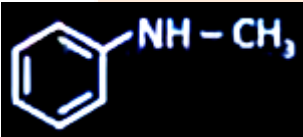
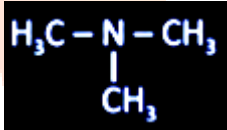
Question: A compound having molecular formula MX_3 has van't hof factor of 2. What is the degree of dissociation?

Options:

- (a) 0.25
- (b) 0.5
- (c) 0.3
- (d) 0.75

Answer: (c)

Question: Which among the following compounds gives positive carbylamine test?

- (A) $CH_3 - NH_2$
- (B) 
- (C) 
- (D) 

Options:

- (a) A and B only
- (b) B and C only
- (c) A and C only
- (d) B and D only

Answer: (a)

Question: Transition metal belonging to 3d series having lowest enthalpy of atomisation in its most stable oxidation state forms oxide MO. Nature of oxide is

Options:

- (a) Highly acidic
- (b) Amphoteric
- (c) Highly basic
- (d) Neutral

Answer: (b)

Question: Given below are two statements.

Assertion (A): Sodium on reaction with alcohols liberates H_2 gas.

Reason (R): Alcohols are acidic in nature.

In the light of the above statements, choose the correct answer from the options given below:

Options:

- (a) Both A and R are correct and R explains A

- (b) Both A and R are correct but R does not explain A.
 (c) A is correct R is incorrect
 (d) A is incorrect R is correct

Answer: (a)

Question: Consider the following statement

Statement-I: D—(+)— Glucose and D—(+)— fructose are formed on hydrolysis of sucrose.

Statement-II: Sucrose is an invert sugar.

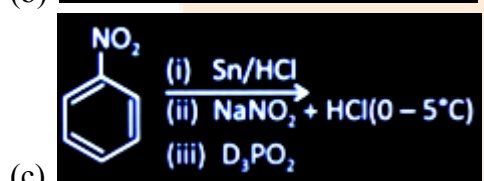
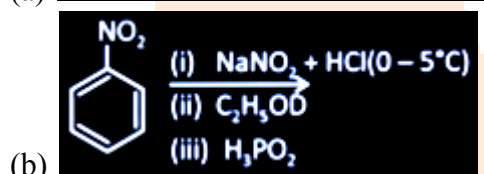
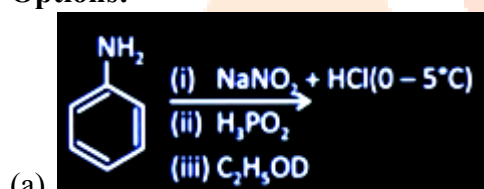
Options:

- (a) Statement-I correct and Statement-II is incorrect
 (b) Statement-I is incorrect and Statement-II is incorrect
 (c) Both Statement-I and Statement-II are correct
 (d) Both Statement-I and Statement-II are incorrect

Answer: (b)

Question: Which one of the following reactions will result in the formation of deuterated benzene (C_6H_5D)?

Options:



(d) None

Answer: (c)

Question: 1 mol of water at 10°C is converted into ice at -10°C . The change in enthalpy for complete conversion is

[Given: C_p of water = $x \text{ JK}^{-1} \text{ mol}^{-1}$]

C_p of ice = $y \text{ JK}^{-1} \text{ mol}^{-1}$

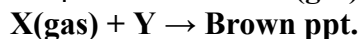
$\Delta H_{\text{fusion}} = z \text{ J}$]

Options:

- (a) $(-10x - 10y - z) \text{ J}$
 (b) $(10x + 10y + z) \text{ J}$
 (c) $(x + y - z) \text{ J}$
 (d) $10(x + y - z) \text{ J}$

Answer: (a)

Question: Consider the following sequence of reaction:



Find out X(gas) and compound Y, respectively.

Options:

- (a) Cl_2 and K_2HgI_4
- (b) NH_3 and K_2HgI_4
- (c) NH_3 and KOH
- (d) HCl and HgI

Answer: (b)

Question: Given below are two statements

Statement-I Reductive ozonolysis of but-2-ene gives ethanal

Statement-II: Reductive ozonolysis of 3, 6-dimethyl oct-4-ene doesn't give compound with chiral carbon.

In the light of the above two statements, choose the most appropriate option.

Options:

- (a) Statement-I correct and Statement-II is incorrect
- (b) Statement-I is incorrect and Statement-II is correct
- (c) Both Statement-I and Statement-II are correct
- (d) Both Statement-I and Statement-II are incorrect

Answer: (a)

Question: Consider the following values of standard reduction potential,

$$E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = +0.34\text{V}, E_{\text{Ag}^+/\text{Ag}}^{\circ} = +0.80\text{V}$$

Find E_{cell} constituted by these two electrodes if, it contains 0.2 M $\text{Ag}^+(\text{aq})$ and 1.5M $\text{Cu}^{2+}(\text{aq})$

Options:

- (a) 0.50 Volts
- (b) -0.50 Volts
- (c) 0.41 Volts
- (d) -0.41 Volts

Answer: (c)

Question: 100 mL HCl solution having pH = 1 is diluted to 250 mL. The pH of final solution is

Options:

- (a) 1
- (b) 1.4
- (c) 1.7
- (d) 2

Answer: (b)

Question: Match the following compounds in List-I with its lone pairs on central atom and bond pairs in List-II.

List-I	List-II
(A) XeF_4	(I) 2 lone pairs and 3 bond pairs

(B) ClI_3	(II) 1 lone pair and 4 bond pairs
(C) SF_4	(III) 3 lone pairs and 2 bond pairs
(D) I_3^-	(iv) 2 lone pairs and 4 bond pairs

Options:

- (a) A-IV, B-II, C-III, D-I
- (b) A-II, B-I, C-III, D-IV
- (c) A-IV, B-I, C-II, D-III
- (d) A-I, B-IV, C-III, D-II

Answer: (c)

Question: Correct order of wavelength of the following colours:

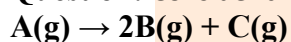
- (I) Red
- (II) Yellow
- (III) Blue
- (IV) Violet

Options:

- (a) $\text{I} > \text{II} > \text{III} > \text{IV}$
- (b) $\text{IV} > \text{III} > \text{II} > \text{I}$
- (c) $\text{II} > \text{I} > \text{IV} > \text{III}$
- (d) $\text{II} > \text{I} > \text{III} > \text{IV}$

Answer: (a)

Question: Consider the following first order reaction



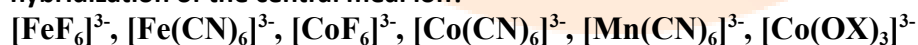
The total pressure at $t = 10 \text{ min}$ is 160 mmHg & $t = \infty$ is 240 mm of Hg. Choose correct statement

Options:

- (a) The reaction will be 100% complete
- (b) $k = 1.693 \text{ s}^{-1}$
- (c) $t = 10 \text{ min}$ $p_A = 40 \text{ mmHg}$
- (d) At $t = 5 \text{ min}$ pressure of system is 180 mmHg

Answer: (c)

Question: How many of the following complex ions are paramagnetic and have d^2sp^3 hybridization of the central metal ion?



Options:

- (a) 2
- (b) 3
- (c) 4
- (d) 5

Answer: (a)

**Question: Statement-I: Di-methyl ether is almost completely soluble in water.
Statement-II Sodium can be used in dry diethyl ether but not ethyl alcohol.**

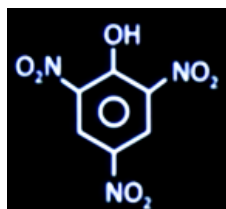
Options:

- (a) Statement-I is correct but statement-II is incorrect
- (b) Statement-I is incorrect but statement-II is correct
- (c) Both Statements are correct
- (d) Both statements are incorrect

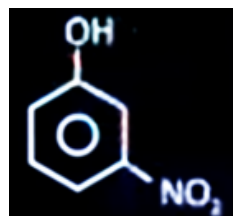
Answer: (c)

Question: Which of the following molecule will not liberate CO_2 gas with NaHCO_3

Options:

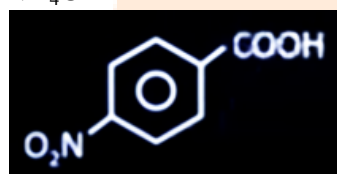


(a)



(b)

(c) NH_4Cl



(d)

Answer: (b)

Question: A 3d-transition series metal oxide in $n+$ oxidation (M^{n+}) state is green coloured oil. The nature of this oxide is

Options:

- (a) Neutral
- (b) Acidic
- (c) Basic
- (d) Amphoteric

Answer: (b)

Question: In air, mass% composition is given as

$\text{N}_2 = 70\%$, $\text{O}_2 = 27\%$, $\text{Ar} = 3\%$

Find the ratio of partial pressure of gases

($\text{N}_2 : \text{O}_2 : \text{Ar}$)

Options:

- (a) 0.73 : 0.02 : 0.01
- (b) 0.67 : 0.31 : 0.09
- (c) 0.67 : 0.31 : 0.03
- (d) 0.73 : 0.25 : 0.2

Answer: (d)

Question: 500 mg of organic compound gives 220 mg of CO_2 . Find mass % of carbon atoms present in organic compound.

Options:

Answer: (12%)



JEE-Main-07-04-2025 (Memory Based)

[MORNING SHIFT]

Maths

Question: The remainder when $64^{64^{64}}$ is divided by 7 is equal to

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (a)

$$64 \equiv 1 \pmod{7}$$

$$64^{64^{64}} \equiv 1 \pmod{7}$$

$$r = 1.$$

Question: Let A be a set defined as $A = \{2, 3, 6, 9\}$. Find number of singular matrices of order 2×2 such that elements are from the set A.

Options:

- (a) 4
- (b) 10
- (c) 12
- (d) 15

Answer: (c)

$$A = \{2, 3, 6, 9\}.$$

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|M| = ad - bc = 0 \Rightarrow ad = bc$$

1. When only one element is used = 4
 2. When exactly two elements are use = ${}^4C_2 \cdot 2 \times 2 = 24$
 3. When exactly three numbers used = 0
 4. When all four used = ${}^4C_1 \cdot 1 \times 2 = 8$
- Total = $4 + 24 + 8 = 36$

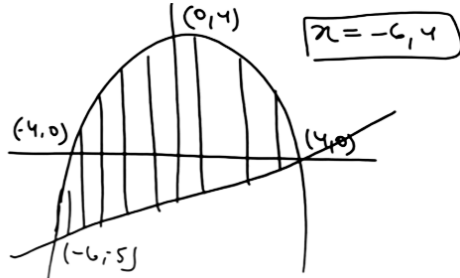
Question: Area bounded by the curves $y = 4 - \frac{x^2}{4}$ and $y = \frac{x-4}{2}$ (in square units) is

Options:

- (a) $\frac{125}{3}$
- (b) $\frac{20}{3}$

- (c) $\frac{80}{3}$
 (d) $\frac{120}{3}$

Answer: (a)



$$\int_{-4}^4 \left(\left(4 - \frac{x^2}{4} \right) - \left(\frac{x-4}{2} \right) \right) dx$$

$$= \frac{125}{3}$$

Question: If x_1, x_2, x_3, x_4 are in GP, then we subtract 2, 4, 7, 8 from x_1, x_2, x_3, x_4

respectively, then the resultant number are in AP then the value of $\frac{1}{24} (x_1, x_2, x_3, x_4)$ is

Options:

- (a) $\frac{2^4}{3^8}$
 (b) $\frac{2^3}{3^9}$
 (c) $\frac{2}{3^9}$
 (d) $\frac{2}{3^8}$

Answer: (b)

$$\Rightarrow x_1, x_2, x_3, x_4 = x_1, x_2r^1, x_1r^2, x_1r^3 = x_1^4r^6 = a^4r^6$$

$$(x_1 - 1), (x_2 - 4), (x_3 - 7), (x_4 - 8) \text{ average in AP}$$

$$\Rightarrow (a - 2)(ar - 4)(ar^2 - 7), (ar^3 - 8)$$

$$(ar - 4) - (a - 2) = (ar^2 - 7) - (ar - 4)$$

$$\Rightarrow ar - a - 2 = ar^2 - ar - 3$$

$$\Rightarrow a(r - 1) - ar(r - 1) = -1$$

$$\Rightarrow a(r - 1)[1 - r] = -1$$

$$\Rightarrow a(r - 1) = 1$$

$$\Rightarrow a(9) = 1$$

$$\boxed{a = \frac{1}{9}}$$

$$ar^2 - 7 - (ar - 4)$$

$$= (ar^3 - 8) - (ar^2 - 7)$$

$$\Rightarrow ar(r - 1) - 3 = ar^3 - ar^2 - 1$$

$$\Rightarrow ar(r - 1) - ar^2(r - 1) = 2$$

$$\Rightarrow ar(r - 1)[1 - r] = 2$$

$$\Rightarrow \frac{ar(r - 1)^2}{a(r - 1)^2} = \frac{-2}{1}$$

$$\boxed{r = -2}$$

$$\frac{1}{24} (a^4r^6)$$

$$= \frac{1}{24} \cdot \frac{1}{9^4} \cdot 2^6$$

Question: If $f(x) = \left\lfloor \frac{x^2}{2} \right\rfloor - \lfloor \sqrt{x} \rfloor \forall x \in [0, 4]$ where $\lfloor . \rfloor$ denotes the greatest integer function, then number of point discontinuity of $f(x)$ is

Options:

- (a) 12
- (b) 8
- (c) 6
- (d) 4

Answer: (b)

$$f(x) = \left[\frac{x^2}{2} \right] - [\sqrt{x}]$$

Check at $x = 0, 1, \sqrt{2}, 2, \sqrt{6}, \sqrt{8}, \sqrt{10}, \sqrt{12}, \sqrt{14}, 4$

Continuous at $x = 0, 4$

Ans = 8

Question: The integral $\int_0^\pi \frac{(x+3)\sin x}{1+3\cos^2 x} dx$ is equal to

Options:

- (a) $\frac{\pi}{3\sqrt{3}}(\pi+6)$
- (b) $\frac{\pi}{2\sqrt{3}}(\pi+4)$
- (c) $\frac{\pi}{\sqrt{3}}(\pi+2)$
- (d) $\frac{\pi}{\sqrt{3}}(\pi+1)$

Answer: (a)

$$I = \int_0^\pi \frac{(x+3)\sin x}{1+\cos^2 x} dx \text{ --- (1)}$$

$$I = \int_0^\pi \frac{((\pi-x)+3)\sin x}{1+3\cos^2 x} dx \text{ --- (2)}$$

$$2I = \int_0^\pi \frac{\sin x}{1+3\cos^2 x} (x+\pi+\pi-x+3) dx$$

$$2I = \left(\int_0^\pi \frac{\sin x}{1+3\cos^2 x} dx \right) (\pi+6) \Rightarrow 2I = \left(\int_0^{\pi/2} \frac{\sin x}{1+3\cos^2 x} dx \right) (\pi+6)$$

let $\cos = T \rightarrow -\sin x dx = dt$

$$I = \left(- \int_1^0 \frac{1}{1+(\sqrt{35})^2} dt \right) (\pi+6)$$

$$I = \left(\int_0^1 \frac{1}{1+(\sqrt{35})^2} dt \right) (\pi+6)$$

$$I = \frac{(x+6)}{\sqrt{3}} \left(\tan^{-1}(\sqrt{35}) \right)_0^1$$

$$= \left(\frac{\pi+6}{\sqrt{3}} \right) \frac{\pi}{3}$$

Question: If α and β are negative real roots of the quadratic equation $x^2 - (p+2)x + (2p+9) = 0$ and $p \in (\alpha, \beta)$. Then the value of $\beta - 2\alpha$ is

Options:

- (a) 11

(b) 13

(c) 7

(d) 5

Answer: (d)

O is greater than both the roots

$$af(0) > 0$$

$$\Rightarrow 1 \cdot 2p + 1 > 0$$

$$\Rightarrow p > -\frac{9}{2}$$

$$D > 0$$

$$(p + 2)^2 - 4(2p + 1) > 0$$

$$p^2 + 4p + 4 - 8p - 36 > 0$$

$$p^2 - 4p - 32 > 0$$

$$(p - 8)(p + 4) > 0$$

$$p > 8 \text{ or } p < -4$$

$$\alpha = -\frac{9}{2}\beta = -4$$

$$\beta - 2\alpha = 5$$

Question: Let the straight line AB : $x + y - 2 = 0$, AC : $3y - x = 2$ intersects x-axis at B and C respectively. If P is the orthocentre of the triangle ABC, then area of the triangle CPB is

Options:

(a) 10 sq. units

(b) 8 sq. units

(c) 6 sq. units

(d) 7 sq. units

Answer: (c)

$$y = \frac{x}{3} + \frac{2}{3}$$

$$CP \equiv (y - 0) = 1(x + 2)$$

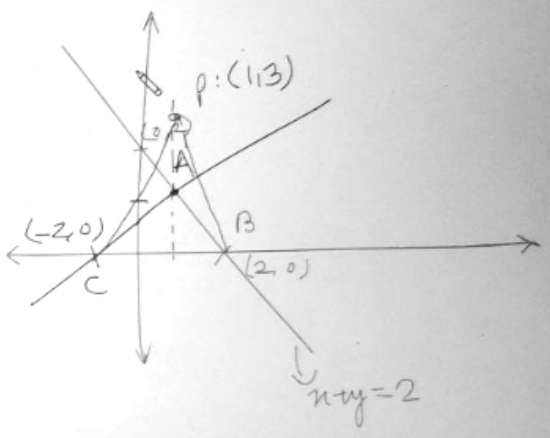
$$BP \equiv (y - 0) = -3(x - 2)$$

$$x + 2 = -3x + 6$$

$$4x = 4$$

$$x = 1, y = 3$$

$$\frac{1}{2} \times 4 \times 3 = 6$$



Question: Consider two statements:

Statement 1 : If $\left(\frac{z+i}{z-i}\right)$ is purely real, then there are exactly 2 complex numbers z .

Statement 2 : If $\left(\frac{z+1}{z-1}\right)$ is purely imaginary, then there are infinite such complex numbers z . Then

Options:

- (a) Statement 1 is true
- (b) Statement 2 is true
- (c) Both statement 1 and statement 2 are true
- (d) Both statement 1 and statement 2 are false

Answer: (b)

I. $\arg\left(\frac{z+i}{z-i}\right) = 0, \pi$

All the complex number lying on $y = x$ is

satisfy it except $-i, i$

II. $\arg\left(\frac{z+1}{z-1}\right) = \pm \frac{\pi}{2}$

Z lies on a circle

Statement 2 is true

Question: If $\vec{a}$ and $\vec{b}$ are unit vectors such that angle between

$\vec{a}$ and $\vec{b}$ is $\sin^{-1}\left(\frac{\sqrt{65}}{9}\right)$ and $\vec{c} = 3\vec{a} + 4\vec{b} + 9(\vec{a} \times \vec{b})$. Then, the value of $\vec{c} \cdot \vec{a} - 3\vec{c} \cdot \vec{b}$ is

Options:

- (a) $\frac{101}{3}$
- (b) $\frac{-101}{9}$
- (c) $\frac{101}{9}$
- (d) $\frac{-101}{3}$

Answer: (b)

$$\vec{c} = 3\vec{a} + 4\vec{b} + a(\vec{a} \times \vec{b}) \text{ take } \vec{a}$$

$$\vec{c} \cdot \vec{a} = 3 + 4\vec{a} \cdot \vec{b}$$

$$\vec{c} \cdot \vec{b} = 3\vec{a} \cdot \vec{b} + 4$$

$$\vec{c} \cdot \vec{b} - 3\vec{c} \cdot \vec{b} = 3 + 4\vec{a} \cdot \vec{b} - 9\vec{a} \cdot \vec{b} - 12$$

$$= -9 - 5\vec{a} \cdot \vec{b}$$

$$\sin \theta = \frac{\sqrt{65}}{9}$$

$$l = \sqrt{9^2 - 65}$$

$$\cos \theta = \frac{4}{9}$$

$$= \sqrt{81 - 65}$$

$$\vec{a} \cdot \vec{b} = \frac{4}{9}$$

$$= \sqrt{16} = 4$$

$$= -9 - 5\frac{4}{9} = \frac{-81-20}{9} = \frac{-101}{9}$$

Question: Let 'P' be the parabola, whose focus is (-2, 1) & directrix is $2x + y + 2 = 0$. Then the sum of the ordinates of the points on P, whose abscissa is -2 is ____.

Options:

(a) $\frac{5}{2}$

(b) $\frac{1}{4}$

(c) $\frac{3}{4}$

(d) $\frac{3}{2}$

Answer: (d)

$$pf = pm$$

$$\sqrt{0 + (k - 1)^2} = \left| \frac{4+k+2}{\sqrt{5}} \right|$$

$$k^2 + 1 - 2k = \frac{k^2 + 4 - 4k}{5}$$

$$5k^2 + 5 - 10k = k^2 + 4 - 4k$$

$$4k^2 - 6k + 1 = 0$$

$$k_1 + k_2 = -\frac{-6}{4} = \frac{3}{2}$$

